MAR BASELIOS COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

QUESTION BANK

23ELB20E OBJECT ORIENTED PROGRAMMING USING JAVA

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MODULE 1

1.	Develop a Java program that stores the marks of students using a Vector. Using appropriate loops and wrapper classes, calculate and display the average marks and the highest mark among the students.	(3)
2.	Develop a Java program to reverse the elements of an array without using another array.	(3)
3.	Given an array of unique integers that were originally sorted in ascending order and then rotated between 1 and n times, develop a program to find and return the minimum element in the array. For example, for nums = [3,4,5,1,2], the minimum element is 1 since the original sorted array was rotated.	(12)
4.	Develop a Java program to find the second largest element in an integer array.	(6)
5.	Develop a Java program to add 10 random numbers between 1 to 100 in a vector. Remove all even numbers from the vector. Display the contents of the vector before and after the removal of even numbers.	(6)
6.	Develop a Java program to swap two numbers without a third variable.	(2)
7.	Develop a Java program to print a Fibonacci sequence using recursion.	(2)
8.	Develop a Java program to find the sum of digits of a given number.	(2)
9.	Develop a Java program to find all duplicate elements in an array.	(6)
10.	Develop a Java program using a lambda expression to add two integers.	(3)
11.	Develop a Java program to find and print the maximum value from a list.	(3)

12.	Identify the output of the following program:
	public class Hello {
	public static void main(String[] args) {
	int x = 5;
	x += 3 * 2;
	System.out.println(x);
	}
	}
		(2)

13.	Develop a Java program to check if a string is a palindrome.	(4)
14.	Develop a Java program to find the average of an array of numbers using constructors and methods.	(6)

15.	Develop a Java program that takes five integers as input and finds the largest and smallest number among them.	(6)
16.	Develop a Java program to check if a vowel is present in a string.	(6)
17.	Develop a Java program that reads an array of integers and separates prime and non-prime numbers into two different lists.	(8)
18.	Develop a Java program to check whether a number is an Armstrong number.	(2)
19.	Develop a Java program that sends notifications via: Email SMS App notification Use interfaces and polymorphism.	(8)

MODULE 2

20.	Develop a Java program to create a simple calculator that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. Use a switch case with proper break statements to evaluate the user's choice of operation and display the result.	(4)
21.	Develop a Java program to create a method that calculates the sum of n numbers using varargs (the ... operator).	(2)
22.	Develop a Java program for a Bank Account Management System using hierarchical inheritance. Create a base class Account that stores common details such as account number and account holder name. Derive two classes SavingsAccount and CurrentAccount from Account. Each derived class should have its own specific data members and methods.	(6)
23.	Develop a Java program to overload the area() method to find the area of a square, rectangle, and circle using their respective formulae: area of square = side × side, area of rectangle = length × breadth, and area of circle = π × radius × radius.	(4)
24.	Develop a Java program to create a Student class with both default and parameterized constructors. Use 'this' keyword to initialize class variables and display the student details.	(4)
25.	Develop a Java program using classes and objects to calculate the average marks of three subjects of three students in your class using methods and constructors.	(4)
26.	Develop a Java program for an Employee Payroll System using multilevel inheritance. Create a base class Employee containing employee ID and name. Derive a class PermanentEmployee that includes basic salary and allowances. Further derive a class Manager that adds bonus and department details.	(6)
27.	Develop a banking system in Java that supports multiple banks. The requirements are Abstract class: BankAccount Methods: deposit(), withdraw() , abstract calculateInterest() Subclasses: SavingsAccount, CurrentAccount.	(6)
28.	Develop a Java program to check if the given number is a prime number.	(2)

29.	Identify whether the below code compile successfully. Provide the output of the following program.
	
	public class Test
	{
	Test() {
	System.out.println("Calling default constructor");
	}
	Test(Test test) {
	System.out.println("Calling parameterized constructor");
		(2)

	}
	public static void main(String[] args)
	{
	Test t = new Test(new Test());
	} }	

30.	Develop a Java program to create a Vehicle class with overloaded constructors and a Car class that inherits from Vehicle and has its own constructor.	(2)
31.	Develop an E-Commerce Payment System using Java. Define an interface Payment with a method processPayment(). Implement it in CreditCardPayment and UPIPayment classes. Create a subclass WalletPayment that extends UPIPayment and overrides processPayment(). Run through the steps of the whole process.	(6)
32.	Develop a Java program to generate a billing system for a shopping mall. The program should: Take the total bill amount and customer type as input (Regular or Premium). Apply a discount using the ternary operator. Regular customers get a 5% discount if the bill is above ₹5000. Premium customers get a 10% discount if the bill is above ₹3000.	(3)
33.	Develop a Java program that prints the numbers from 1 to 100. But for multiples of three, print "Fizz" instead of the number, and for the multiples of five, print "Buzz". For numbers that are multiples of both three and five, print "FizzBuzz".	(3)
34.	Develop a Java program to find the sum of two complex numbers. Use suitable constructors and methods.	(6)
35.	Develop a Java program to implement a student management system that allows users to add, delete, search, and display student records using constructors and methods.	(6)
36.	Build a Shopping Cart System using Java where: An abstract class Product has methods calculateDiscount() and displayProduct(). Subclasses Electronics and Clothing override calculateDiscount(). A method calculateTotal(int... prices) uses varargs to find the total cart value dynamically.	(6)
37.	Develop a Java program to create a library management program. Create an abstract class Book with methods getFine() and displayBook(). Subclasses Fiction and NonFiction override getFine() (e.g., Fiction books have a late fine of Rs.1 per day, NonFiction books have Rs.2 per day).	(6)
38.	Develop a Java program to maintain product inventory: Add products with quantity, Update stock, Alert when stock goes below threshold.	(8)
39.	Develop a Java program to read student data from a CSV file: Calculate average marks, Display top scorer.	(8)
40.	Design a Java program with a base class Member containing name and membership ID. Derive StudentMember and FacultyMember classes. Each	(6)

	derived class should calculate membership fee differently using method overriding.	
41.	Create a class Course with course name and duration. Create a subclass PaidCourse that adds course fee and discount. Use constructors and super keyword to initialize values.	(4)
42.	Develop a Java program using constructor overloading to handle prepaid and postpaid recharge plans with different parameters.	(4)
43.		
44.		
45.		

MODULE 3

46.	With suitable examples, explain how Java exception handling avoids runtime failures and ensures graceful error recovery.	(2)
47.	Explain the difference between an error and an exception.	(2)
48.	Explain the meaning of throwing an exception with a simple example.	(2)
49.	Explain the type of exception handling mechanisms with appropriate examples	(6)
50.	Explain the steps involved in creating a user-defined package in Java. Write a program to create a package and access a class from another package.	(6)
51.	List any four Java's Checked Exceptions.	(2)
52.	Explain the difference between Throw and Throws in Java.	(2)
53.	Write a Java program that throws a user-defined exception InsufficientBalanceException when withdrawal amount exceeds balance.	(6)
54.	Develop a Java program that handles exceptions when a file name is empty or file size exceeds the limit using throw.	(4)
55.	Create a program that validates student marks. If marks are less than 0 or greater than 100, raise an exception and handle it gracefully.	(4)
56.	Define a user-defined exception InvalidLoginException. Throw it when username or password does not match predefined values.	(4)

MODULE 4

57.	Develop a Java program that takes a user's full name as input and displays the name in uppercase, lowercase, and its length.	(2)
58.	Develop a Java program to convert the number “0” in the String “C0unter” to the letter “o”.	(2)
59.	Develop a Java program to create and start a thread.	(2)
60.	Develop a text analyzer using Java that takes a string input and performs the following operations: finds its length, extracts specific characters, modifies a substring, and converts numerical values to strings.	(6)
61.	Develop a username validator using Java that checks whether a given username meets the following criteria: At least 6 characters long, contains only letters and digits, Starts with a letter.	(6)
62.	Develop a multithreaded Java program with two threads. One thread generates evennumbersfrom1 to 100 and another thread generates odd numbers from1 to 100. Ensure that the threads do not interfere.	(6)
63.	Develop a Java program that creates an ArrayList of names and a HashMap with names and ages. Print the contents of both the ArrayList and the HashMap. Implement a method to check if a specific name exists in both collections.	(6)
64.	Develop a thread-based railway reservation system using Java where multiple users (threads) try to book train tickets simultaneously, ensuring proper synchronization to prevent overbooking.	(6)
65.	Develop a password strength checker using Java that verifies if a given password contains uppercase, lowercase, digits, special characters, and has a minimum length using string handling methods.	(6)
66.	Develop a Java program where: Students can enroll in multiple courses Prevent duplicate enrollment Display registered courses per student.	(6)
67.	Write a Java program that extracts username and domain name from an email ID and displays them separately using string methods.	(4)
68.	Develop a program that counts how many times each word appears in a sentence using HashMap.	(4)

69.	Use ArrayList to store student names and attendance percentage. Display students with attendance below 75%.	(4)
70.	Develop a multithreaded Java program where two threads increment a shared counter. Use synchronization to avoid incorrect counts.	(6)
71.	Write a Java program that removes duplicate characters from a string and displays the compressed version.	(4)
72.	Store website URLs accessed by users in an ArrayList. Display unique URLs using Set.	(4)
73.	Develop a Java program that tracks website page visits: Store URL and number of visits Display most visited page.	(6)

MODULE 5

74.	Develop a Java program using Swing to create a frame and a text box.	(2)
75.	Develop a Java program to create a frame and a button named “Submit” in it.	(4)
76.	Develop a student registration form using Swing with fields for Name, Age, Email, and Course. The details must be stored in the MySQL Database. Display the entered details in the terminal when the user submits and show a submitted successfully pop-up.	(6)
77.	Develop a JDBC-based student database that allows adding, deleting, and updating student records using Swing forms and MySQL database.	(6)
78.	Develop a java GUI Program to calculate the factorial of a number.	(6)
79.	Develop a Java program using JDBC to connect to a MySQL database. The program should insert a new record into a student’s table with fields id, name, and age, and then display all the records from the table.	(6)
80.	Build a restaurant ordering system using Java where users can select food items, and the total bill is stored in a database.	(6)
81.	Develop an employee registration form using Swing with fields for Name, Age, Email, and Department. The details must be stored in the MySQL Database. Display the entered details in the terminal when the user submits.	(6)
82.	Design a Java GUI to book doctor appointments and store patient details in database.	(8)
83.	Create a GUI application that calculates gross salary and stores employee details using JDBC.	(8)
84.	Develop a Faculty registration form using Swing with fields for Name, Age, Email, and Department. The details must be stored in the MySQL Database. Display the entered details in the terminal when the user submits.	(8)
85.	Develop a Java program that creates a GUI application for a temperature converter. The program should allow users to convert temperatures between Celsius and Fahrenheit. Provide its sample output.	(8)
86.	Develop a Java program that creates a GUI application for a password validator. The program should allow users to input a password and check if it meets the following criteria. i. Window with password input, "Check Password" button, result label. Validate password: ≥8 chars, ≥1 uppercase, ≥1 lowercase, ≥1 digit.	(8)